



Huggingface

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HuggingFace

HuggingFace (HF) is a platform + ecosystem for **building, sharing, training, deploying, and experimenting with machine learning models**, especially **NLP and LLMs**.

Think of it as:

“The GitHub + AppStore of AI Models.”

It provides:

A public **Model Hub**

A **Dataset Hub**

A **Spaces** hub to host demo apps

A Python library called **Transformers**

Tools for training, evaluating, and deploying models



HuggingFace Model Hub

A huge repository of **20,00,000+** models and growing...

- **NLP models**
 - BERT, RoBERTa, DistilBERT
 - GPT-2, BLOOM, LLaMA, Mistral, Falcon
 - T5, BART
 - Whisper (speech-to-text)
 - LayoutLM (documents)
- **Vision models**
 - ViT, CLIP
 - YOLO, DETR
 - SAM (Segment Anything)
 - Stable Diffusion (image generation)
- **Audio models**
 - Whisper
 - AudioLM
 - MusicGen
- **Multimodal**
 - LLaVA
 - Qwen-VL
 - CLIP



HuggingFace Model Hub

Class	What it loads	For which models
AutoModel	Base transformer, no specific task	Encoder/decoder-only layers
AutoModelForCausalLM	Decoder-only GPT models	Text generation
AutoModelForMaskedLM	BERT-style, Encoder-only masked models	MLM tasks
AutoModelForSeq2SeqLM	Enc-Dec models	Translation/summarization
T5ForConditionalGeneration	T5-specific, Conditionalgen i.e. model gives output text that depends on that input	Any text → text
StableDiffusionPipeline	Diffusion models	Image generation



HuggingFace Datasets Hub

A huge repository of **100,000+ datasets** for ML and growing...

You load datasets with 1 line:

```
from datasets import load_dataset  
dataset = load_dataset("imdb")
```

Supports:

- Text
- Audio
- Image
- Video
- Multimodal
- Huggingface hosting your own datasets



HuggingFace Access Token

Model Type	Token Needed?	Why
Public models	✗ No	Fully free
License-restricted	✓ Yes	Must accept license
Request-access models	✓ Yes	Controlled release
Already cached	✗ No	Token stored locally

You must select the

“Make calls to Inference Providers” to access the huggingfaceendpoint

“Read access to contents of all repos under your personal namespace”

“Read access to contents of all public gated repos you can access “

To access the HuggingFace opensource models and must accept terms and conditions for some models



HuggingFace Features

Component	What It Is	Key Features	Typical Use
HuggingFace Spaces	Free app hosting platform	Streamlit, Gradio, Docker, Static Apps	Deploy demos (chatbots, LLMs, image generators)
Transformers Library	Main HF library for models	Pretrained models, tokenizers, pipelines, training utils	Load, train, and use NLP/vision/audio models
Tokenizers Library	Fast Rust-based tokenizer engine	BPE, WordPiece, SentencePiece, Unigram	Build or load tokenizers for LLMs
Accelerate	Simple distributed training framework	Multi-GPU, TPU, multi-node, mixed precision	Train LLMs efficiently at scale
Trainer API	High-level training loop	LoRA, QLoRA, PEFT, FP16/BF16	Fine-tune transformer models easily
Inference Endpoints	Managed model hosting as API	Private endpoints, GPUs, autoscaling	Deploy models like an OpenAI-style API
Text Generation Inference (TGI)	LLM inference server	Fast generation, quantization, optimized for big models	Production deployment of models like Falcon/Mistral/LLaMA
Safetensors	Model weight storage format	Safe, fast, zero-copy, no pickle	Load/save transformer weights securely

Thank you